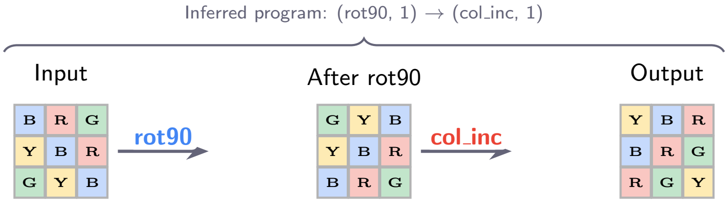
ProgramEncoder infers composition from $K=3$ support pairs (input $\rightarrow$ output examples)



Algebraic composition in HDC space:

$$\underbrace{z_{\text{out}}}_{\text{output}} = \underbrace{\text{col\_inc}}_{\text{colour LUT}} \circ \underbrace{\text{rot90}}_{\text{cell permutation}} (z_{\text{in}}) \equiv P_{\text{rot}} \cdot z_{\text{in}} \text{ then } \text{LUT}[\text{colour} + 1]$$

Out-of-distribution generalisation (zero-shot, no retraining):

Novel rule	**Composition**	**Accuracy**
rotate_270	rot90^3	100%
recolour_dec	col_inc^9	100%
flip_vertical	$\text{rot90}^2 \circ \text{reflect}$	100%
shift_down	$\text{rot90} \circ \text{translate} \circ \text{rot90}^3$	100%